VISHWAS · SIH26079 · TEAM HELLFIRE LABS

Mahika

Stakeholder interviews and pitch

Final prep — script, judge Q&A, and every number traced

BBA / BA

YOU LEAD. Slide 1 (open) · bridge into slide 4 · Slide 5 (first half) · the close · and you run Q&A

about 2 minutes of speaking across four appearances, plus Q&A traffic control

LEADS THE PRESENTATION

YOUR JOB

You are leading this presentation. That means three jobs beyond your own content: you set the register in the first fifteen seconds, you carry the thread between five other speakers so it reads as one argument rather than six reports, and in Q&A you decide who answers. Your own material — the field research — is the human half of the pitch, and your discipline about what three interviews can and cannot prove is itself part of why the judges should believe the rest.

1 RUNNING ORDER

Mahika leads. Your rows are highlighted.

#	SPEAKER	WHAT HAPPENS	TIME
1	**Mahika**	**Slide 1 — opens. Team, problem statement, the hook.**	**~40s**
2	Arnav	Slide 2 — what VISHWAS is, and why the forecast underneath decays.	~70s
3	Devansh	Slide 2 — the measurement, and the result.	~60s
4	Ankit	Slide 3 — opens it: what an ensemble is, and why we score against rain gauges.	~35s
5	Devansh	Slide 3 — the pipeline, the stack, features, label, model, the map.	~75s
6	**Mahika**	**Bridge into slide 4.**	**~10s**
7	Paras	Slide 4 — feasibility, the numbers, the three defects.	~65s
8	Arnav	Slide 4 — viability, the four risks, the precision choice.	~40s
9	**Mahika**	**Slide 5 — the three interviews.**	**~45s**
10	Dhruv	Slide 5 — who it serves, MoES's numbers, the benefits. Then slide 6 — sources.	~75s
11	Devansh	Slide 6 — our own measurement.	~20s
12	**Mahika**	**Closes. The tagline, then into questions.**	**~20s**

2 THE STORY — EVERYONE MUST BE ABLE TO SAY THIS

If a judge stops you in the corridor. Thirty seconds. Learn the shape, not the words.

> Every weather forecast India issues arrives with no honesty label.
>
> Forecasters judge confidence today by ensemble spread — run the model 11 times from slightly different starting conditions, and if the runs disagree, distrust the forecast. We measured that method against India's own rain gauges, across all four seasons and three years.
>
> Compared crudely it looks strong: ensemble spread appears to explain 43.9% of the variance in forecast error. Compared fairly — within the same rainfall conditions — it explains under 1%.
>
> So we built the replacement. On that same fair comparison it scores 0.565 and VISHWAS scores 0.728, on a test year the model had never seen.
>
> ***Most teams pitch a hypothesis. We pitch a measurement.***

3 LEADING THE ROOM

You are running the room, not just speaking in it. These are the parts of the job that are not on any slide.

BEFORE YOU START

- Say the team name and the problem statement number clearly and slowly — the judges are matching you against a list, and half the teams garble this.
- Do not read the slide. Everything on slide 1 is already in front of them; your forty seconds are for the one thing that is not written down, which is what the project is for.
- Name the disciplines. Six people across B.Tech, BCA and BBA is unusual and it is worth ten seconds.

CARRYING THE THREAD

- You appear four times. Each appearance exists to make the next speaker land, not to add content.
- Hand over by name and by subject: “Ankit will show you how it is built” beats “over to Ankit”. The judges are tracking who owns what — help them.
- The bridge into Paras is the one that matters. “Paras is going to tell you how we know it works, including the three things we got wrong” turns a defects list into a credibility claim before he has said a word.

IF A SPEAKER OVERRUNS OR FREEZES

- Overrun: let them finish the sentence, then step in with the next handoff line. Do not signal at them in front of the judges.
- Freeze: take the next line yourself if you know it, or go straight to the handoff. “—and Devansh will take the numbers.” Nobody in the audience knows what was supposed to happen.
- You are the only person who should be watching the clock. Everyone else should be watching the judges.

RUNNING THE QUESTIONS

- Repeat or compress the question before anyone answers it. It buys the answerer four seconds and it makes sure the room heard it.
- Route it by name using the table below, then stop talking. Do not answer technical questions yourself and do not add to a teammate's answer unless they hand back to you.
- If two judges ask at once, take them in order and say so: “Let me take that one first, then come back to yours.”
- If an answer is going long, close it: “—and we can go deeper on that afterwards if it is useful.”

- If nobody knows: “We do not have that to hand and we would rather not guess — can we come back to you?” That is a legitimate answer for this team, and given that we removed one of our own figures rather than present a number we could not trace, it is a consistent one.

THE ONE QUESTION YOU SHOULD TAKE YOURSELF

- “Why this team?” is yours. Six people, three disciplines, and every figure traced or removed. Nobody else should answer it.

WHO ANSWERS WHAT

IF THE QUESTION IS ABOUT…	SEND IT TO
Model, score, AUC, features, SHAP, calibration, why not a neural network	Devansh
Data source, fetch, GRIB, S3, why Wednesdays, domain size, the pipeline	Devansh
What an ensemble is, why IMD gauges rather than reanalysis	Ankit
A real example of a bust, Mumbai 2005, Kerala 2018, how bad the forecast is	Arnav
Metrics, POD/FAR/CSI/ETS, alert budget, leakage, the three defects, statistics	Paras
Viability, the risks, why GEFS and not NEPS	Arnav
The interviews, users, the pitch, why this team	Mahika — you
Sources, the MoES numbers, impact, cost, roadmap, the dashboard gap	Dhruv

4 WHAT CHANGED IN YOUR SCRIPT

Your wording is your own. Only these points moved.

- CUE CHANGE — Arnav now speaks between Paras and you. Paras's script was too long for him, so viability and the four risks moved to Arnav, who closes Slide 4. Your bridge into Paras is unchanged, but your cue into Slide 5 is now Arnav's line, not Paras's.
- Your “Well, yes!” is replaced. As written it answered “Can we actually trust this forecast?” with yes — the opposite of your point, in the first twenty seconds.
- The bridge into Paras now names the three defects instead of “challenges we identified and addressed.” Calling them defects before he speaks is what turns the list into a credibility claim.
- Only you greet the judges. The greetings have been removed from Arnav's and Dhruv's scripts — there were three in the deck, and two of them said a different time of day to yours.
- Two optional trims marked in Slide 5. At full pace the whole presentation runs about 9m45 against the 8-minute plan, and your section is the longest.
- NEW — the open now states the problem statement itself, not just its number: the title, plus two short sentences on what the Ministry described and what it asked for. It was the one thing missing from an otherwise complete open, and it is the first thing a

MoES judge checks. Trimmed to about 15 seconds — the long version listed five weather systems and the method; three systems and the ask carry it.

5 YOUR SCRIPT

Grey boxes are stage directions — do not say them. Lines tagged TRIM are safe to cut if you are running long.

SLIDE 1 — THE OPEN

▸ *If your slot is after 12, say "Good afternoon"; after 4, "Good evening." You are the only one who greets.*

Good morning, respected judges.

We are Team Hellfire Labs, presenting VISHWAS for Problem Statement SIH26079 — "AI-Based Forecast Bust Detection for Medium-Range Weather Forecasts" — under the Ministry of Earth Sciences and NCMRWF.

▸ *Number and title together — the judges are matching you against a list.*

The Ministry's problem is this. Medium-range forecasts sometimes fail badly during monsoon depressions, heavy rainfall and cyclones. They call those forecast busts, and operational decisions get made on them anyway.

What they asked for is an AI system that flags where, and how far ahead, a forecast is likely to be wrong. A forecast confidence indicator.

▸ *Portal's own wording, compressed. If MoES questions the framing, it is theirs. IF ASKED how: "by comparing current forecast patterns against historical forecast error behaviour" — also their sentence.*

VISHWAS means confidence — and that is exactly the problem we are addressing.

We receive weather forecasts every day. But before making an important decision based on one, a critical question remains:

Can we actually trust this forecast?

▸ *PAUSE — let it sit for a beat.*

Today, there is no answer to that question. Every forecast arrives with no honesty label.

VISHWAS does not create or correct a weather forecast. Instead, it predicts when an existing forecast could be significantly wrong.

Our team brings together six members from B.Tech, BCA and BBA — combining technical development, analytical evaluation, research, and stakeholder understanding.

To begin, Arnav will explain what VISHWAS is, and why today's forecasting system needs a reliability layer.

BRIDGE — AFTER SLIDE 3, INTO SLIDE 4

▸ *Devansh stops on the ablation and looks at you. Step in before Paras.*

So, that is how VISHWAS is built.

But building a model is not enough. We also need to know whether it actually works.

Paras will now take you through the feasibility, the performance evaluation — and the three defects we found in our own work, and fixed.

▸ *That last line is the most important sentence you say. Do not rush it, and do not soften "defects".*

SLIDE 5 — STAKEHOLDER VALIDATION

▸ *CUE CHANGED. Paras no longer hands to you — Arnav does. He now closes Slide 4 with viability and the risks, and ends on "Ab Mahika aapko batayengi ki humne field mein kya suna." That is your cue. Do not step in when Paras stops.*

We did not want to validate this problem only through data and models.

Hum yeh bhi samajhna chahte the ki jin logon ko weather forecasts ke basis par decisions lene padte hain, unke liye forecast reliability kitni relevant hai.

So, in August, we conducted three exploratory stakeholder conversations.

First, we spoke with a farmer in Assam. Unhone share kiya ki expected rainfall ke basis par unhe apni sowing schedule shift karni padi — but eventually, woh rain nahi hui.

TRIM The impact was not just an incorrect prediction — seeds, labour, time, and ultimately money, sab involved tha.

When we discussed a reliability signal alongside the forecast, he said that if properly validated, he would consider using it while making decisions.

Second, we spoke with a representative from the Krishi Vibhag in Unnao, Uttar Pradesh. They explained that routine planning relies on statistical norms and available information.

When we asked whether advance information about forecast reliability could help with planning, their response was —

"Very, very beneficial."

Third, we spoke with an NDRF constable in Guwahati. He shared instances where preparedness and mobilisation were carried out for forecast heavy rainfall — but the predicted event did not occur.

TRIM Unka perspective clear tha — identifying possible forecast failure earlier could support better preparedness and more efficient resource planning.

Now, I want to be very precise here —

Yeh three conversations hain, survey nahi.

We are not converting these responses into percentages, or claiming that these three stakeholders represent every user.

But despite representing three very different positions — a farmer, an agricultural department representative, and an emergency-response professional — ek common gap repeatedly saamne aaya.

People don't just need to know what the forecast says. They also need to know when that forecast may not deserve the same level of trust.

And with that, Dhruv will now take you through who VISHWAS can serve, and the broader impact of addressing this gap.

THE CLOSE — AFTER SLIDE 6

▸ *Devansh stops on "...make the comparison fair" and looks at you. Stand where you opened.*

That is VISHWAS.

Six people. Three disciplines. One common objective:

To make weather-based decisions more informed — by not only asking what the forecast says, but how much confidence we should place in it.

Every figure presented today has been traced to a government source, peer-reviewed research, or our own documented analysis — and where we could not confidently support a figure, we chose to remove it.

Because in a system built around reliability, our own evidence must also be reliable.

So our message is simple:

Don't just know the forecast. Know whether to trust it.

Thank you. We would be happy to take your questions.

▸ *Then stop talking. Let the first question come — do not fill the silence.*

6 EVERY NUMBER YOU SAY, AND WHERE IT COMES FROM

If a judge challenges a figure, give the source — not a defence.

CLAIM	SOURCE / WHAT IT REALLY MEANS
SIH26079 / Smart Automation / Software	Confirmed against the official portal listing for this problem statement.
Ministry of Earth Sciences / NCMRWF	The organisation and department named on the problem statement.
three conversations, August 2026	Your own field research. Qualitative, exploratory, each attributed to one named role.
'very, very beneficial'	A direct quote from the Krishi Vibhag conversation — the only direct quote on the deck. Say it as a quote.

7 QUESTIONS THE JUDGES WILL ASK YOU

Yours to answer. Read them out loud once — an answer you have said aloud comes back under pressure; one you have only read does not.

Q1 **Only three interviews? That is not a sample.**

Correct, and we do not present it as one. These are three exploratory qualitative conversations, each attributed to one named role, and we deliberately do not convert them to percentages.

What they establish is that the problem is felt in the field, by people in different positions — a farmer, an agriculture department, a rescue service. What they do not do is measure how many people feel it, and we do not claim they do.

The measurement in this project is the meteorological one, and it is on the previous slide.

⚠ If someone presses further, do not get defensive. Agreeing with the limitation IS the answer.

Q2 Would a forecaster actually use this?

Our primary user is the duty forecaster at a centre like NCMRWF or IMD, where confidence is assessed by hand on every cycle — that is the person whose judgement we are automating, and running this in front of one of them is the next thing we would want to do.

The three people I spoke to are downstream. They act on the forecast rather than issue it, and they are the ones who bear the cost when it is wrong.

Q3 Did they understand what you were describing?

I described it as a reliability label on the forecast they already receive, not a new forecast.

The farmer's response is the one I would flag as honest rather than enthusiastic — he said he would act on it once it had been validated. He did not say he would trust it on day one, and I think that is the right instinct to have about a new system.

Q4 How did you find them, and how did you conduct the conversations?

Direct outreach in August 2026, across Assam and Uttar Pradesh, in the language each person was comfortable in. I kept the questions open — what do you do with the forecast, what happens when it is wrong — rather than describing the product first and asking whether they wanted it.

⚠ FILL THIS IN with your own detail before Thursday: how you reached each person, roughly how long each conversation ran. Be specific and do not over-claim.

Q5 Why should the ministry care?

Because the ministry's own numbers say the problem is real. IMD's heavy-rainfall warning accuracy runs from 85% at Day 1 down to 58% at Day 5 — that is MoES's own answer to Lok Sabha Question 2953, in December 2025.

Skill falls with lead time, and that is exactly the window VISHWAS scores. Dhruv can take you through the sourcing.

Q6 What is your pitch in one line?

Don't just know the forecast. Know whether to trust it.

Q7 Why is this team the right team for it?

Six people across three disciplines, and the split is real rather than decorative. The B.Tech side built the pipeline and the model; BCA carried the failure case studies and the evaluation integrity; and the BBA and BA side did the field research and the sourcing — which is why every figure on this deck traces to a government source, a peer-reviewed paper, or our own logs.

The clearest evidence of that is a number we took off: we had a false-alarm ratio in an earlier draft, could not trace it to its table, and removed it rather than present it. Dhruv can tell you about that one.

Q8 Is there anything similar already?

Ensemble spread is the incumbent, and it is what forecasters use today — so we treated it as the thing to beat rather than pretending there was nothing there. The contribution is that we measured it properly first, and it turned out to carry almost no information once you compare like with like.

8 IF IT IS NOT YOURS, HAND IT OVER

Naming the right teammate reads as a team that knows its own work. Guessing reads as one that does not.

IF THEY ASK ABOUT…	SAY THE NAME
Anything about the model, the score, or the method	Devansh
The data pipeline and the fetch	Devansh
What an ensemble is, why gauges rather than reanalysis	Ankit
A documented bust example	Arnav
Metrics and the defects we found	Paras
Viability, the risks, why GEFS and not NEPS	Arnav
Sources, the MoES numbers, impact, roadmap	Dhruv

SITUATION	WHAT TO SAY
You don't know the answer	“I don't have that number to hand and I'd rather not guess — can I come back to you on it?” That is a strength for this team, not a weakness: when we could not trace one of our own figures, we removed it rather than present it.
It's a teammate's area	“That's <name>'s part — <name>, do you want to take that?” Say the name. Don't answer for them.
A judge challenges a number	Give the source, not a defence. Every figure on the deck traces to a government source, a peer-reviewed paper, or our own logs.
A judge says 'you're just predicting heavy rain'	Anyone can answer this one: we removed the rainfall forecast from the model entirely and the score went UP, 0.728 to 0.731.

9 NEVER SAY THIS

Every one of these was a real error caught during the build, or a rule set deliberately.

- × Never say '43.9% of forecast error'. It is 43.9% of the VARIANCE in forecast error. This is the single easiest thing to get wrong under pressure.
- × Never quote the margin as '+0.163'. Say '+0.13 to +0.16'.
- × Never say 'deciles' or 'the wettest tenth' about the model result. It is EIGHT bins. (Deciles are correct only for the separate Phase 1 correlation analysis.)
- × Never quote any accuracy figure. Base rate is about 6%, so accuracy is meaningless — and quoting it signals a team that does not understand imbalance.
- × Never say 'AI-adjusted forecast' or imply we correct the forecast. We DETECT busts. We do not correct anything.
- × Never say the pooled ROC-AUC pair (0.909 vs 0.885) — not on a slide, not in a figure, not aloud, not even if a judge asks directly. Answer with the pooled MARGIN, +0.023, which is published on slide 4. Quoting the pair as performance would commit, in our own work, exactly the error our pitch accuses the incumbent of.
- × Never quote +0.663 as 'the baseline we beat', and never quote a margin range containing +0.035 (that one came from a filter conditioning on the outcome).
- × Never call the regions 'IMD official regions'. They are approximate lat/lon boxes.
- × Never present the confidence map's date as evidence of general performance. It is ONE date, illustrating the form of the product.
- × Never convert the three field interviews into percentages or 'N% of users said'. They are three exploratory conversations.
- × Never quote 98%→46%, ₹13,331 cr, '50% faster', '~30% improvement', or a 0.34–0.37 false-alarm ratio. All were in earlier drafts; none could be traced; all were removed. Do not say any of these numbers out loud even while explaining that you removed them.
- × Never say SHAP and permutation importance 'agree on the drivers'. They pick the same top three in a different order.

10 NUMBER CHEAT SHEET

Same in all six packs. If two of you quote a figure differently in the same room, that is the damage.

WHAT	NUMBER	WATCH OUT
Spread explains, pooled	43.9% of the VARIANCE in forecast error	not '43.9% of forecast error'
Spread explains, fair comparison	0.17%–0.90% → say 'under 1%'	never quote +0.041 alone
VISHWAS score (stratified ROC-AUC)	0.728	—
Ensemble spread score	0.565	—
Margin	+0.13 to +0.16 · 95% CI +0.108 to +0.193	never '+0.163' alone
Matched 10% alert budget	VISHWAS catches 64% of busts vs 55%	—
POD / FAR / CSI / ETS at 10%	0.641 / 0.586 / 0.336 / 0.300 vs 0.546 / 0.647 / 0.273 / 0.235	—
Features	23, all from GEFS, none from the truth	not 24
Ensemble members	11 (control + 10 perturbed)	—

WHAT	NUMBER	WATCH OUT
Domain	1,252 gauge points of a 5,041-point analysis box (24.8%)	the 5,041 box is land AND ocean; only the 1,252 is land
Data fetched	50.3 GB · 104 forecast dates (2018-2019)	this study only; both studies together are 124.5 GB over 256 dates
Labelled rows	1,302,080 in this study · 651,040 in the test set	don't conflate the two; across both studies it is 3,205,120
Five-year retest	margin +0.149, 95% CI +0.116 to +0.203, 104 forecast dates	quote the MARGIN only, never an AUC beside 0.728
Live cycle scored	GEFS 20260907 and 20260908, ~100 s, ~210 MB, CPU	the live run is DONE — never call it future work
High-risk share	7.78% gauge-verified · 14.06% unverified part of the shipped map	ALWAYS name the domain; never blend them
Train / test	trained on 2018 · tested on all 52 Wednesdays of 2019	—
Base rate	about 6%	so never quote accuracy
Bust definition	|forecast − observed| above the 95th percentile, per region and lead time, threshold set on training data only	—
Rainfall bins	EIGHT bins · 40.92% bust rate in the wettest, 0.15% in the driest	never 'deciles' or 'wettest tenth'
Ablations	full 0.728 · no month 0.722 · no rainfall forecast 0.731 · no 850 shear 0.726 · spread-only 0.562	—
Underlying GEFS, heavy rain (≥64.5 mm)	detects 23% of events at Day 1, 0.2% at Day 10	this is the forecast, NOT our model
IMD warning accuracy (MoES)	85, 73, 67, 63, 58% for Day 1–5	Lok Sabha Q.2953, 17 Dec 2025
The tie	ensemble spread is exactly 0.00 mm on 189,758 test rows — a third of the set	—
Calibration	Brier 0.04217 vs 0.04749 · resolution 1.4× the incumbent's	—

ONE LAST THING

The strongest thing about this project is not the score. It is that we measured the method used today before proposing a replacement, and that we removed our own figures when we could not source them. If you land one impression, land that one.